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ORIGINAL ARTICLE

Using machine learning method to predict food waste in catering industry under high resolution: a case in Dongguan

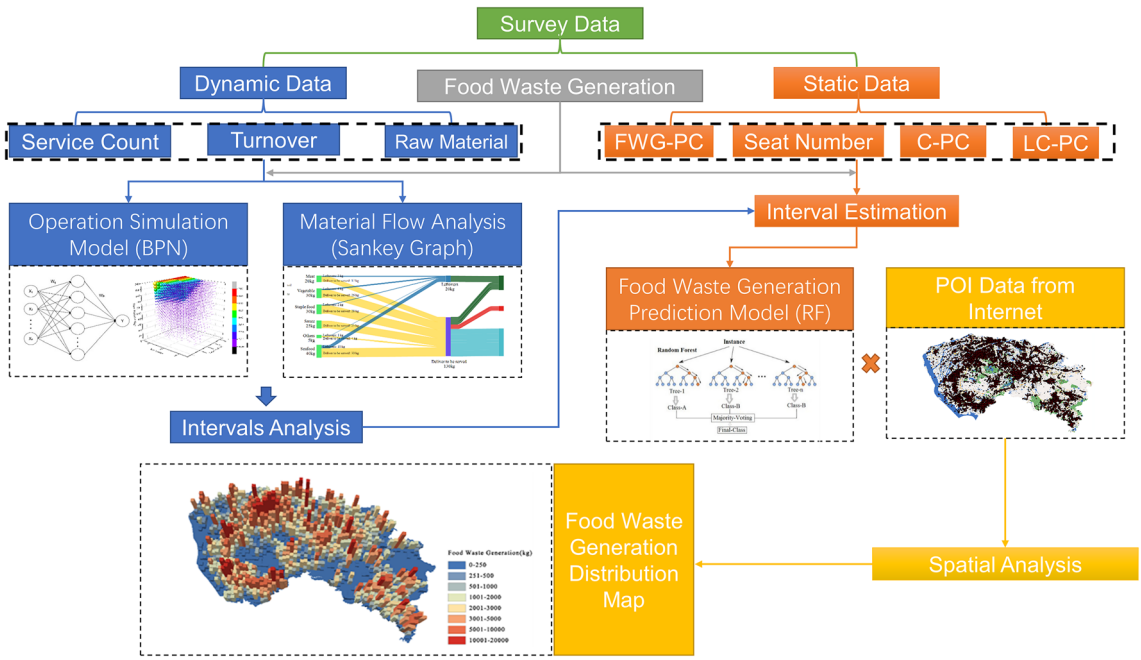
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Abstract

Waste classification is comprehensively carried out in China as an important national-level policy, and the large amount and the wide range of food waste generation (FWG) cause problems in the collection, transportation, and treatment. This study has conducted the prediction of FWG from the catering industry under high resolution, and provided suggestions and insights for food waste management and treatment. Taking Dongguan as an example, a Back Propagation Network (BPN) model is used to predict FWG under different operation data, and based on the acquired theoretical FWG numerical distribution, the intervals used to divide FWG values are determined. Then a Random Forest (RF) model is applied to predict the FWG intervals of the restaurants in the Point of Interest (POI) data. FWG of 96,303 restaurants is predicted, and the predicted FWG from the catering industry is about 3,106 t per day. Variation of FWG in different categories of restaurants, the material flow of FWG at the restaurant level, patterns of FWG at the restaurant level, and spatial patterns of FWG at the city level are also investigated. Suggestions for improvement of food waste collection standard and source reduction of FWG, and insights into food waste collection and distributed treatment system are raised.

Graphical abstract



Keywords Food waste · Geographic information system · Big data · Spatial analysis · High-resolution urban grid

Extended author information available on the last page of the article

Introduction

Garbage siege has become a serious problem for further high-quality development of socioeconomic in China, as the population explosively grows and reaches 1.4 billion in 2020 [1]. According to Urban Construction Statistical Yearbook, the delivering quantity of domestic waste in China has increased at an increasing rate of about 6% since 2016, and reached over 242 million tons in 2019 [2]. The increasing domestic waste generation causes a significant gap between the delivery quantity and the harmless treatment capacity. In order to better realize the reduction and utilization of solid waste, waste classification is comprehensively carried out in China as an important national-level policy [3], and food waste is the major part of municipal waste, accounting for the share of 56.79% reportedly [4]. Because the total amount of food waste generation (FWG) is large, but the generation sources are scattered within a broad geographic spatial scale, the pattern of FWG is highly complicated, resulting in chaos of food waste collection, then a low recycle rate of food waste.

Improving the recycle rate of food waste can accelerate the achievement of Carbon Peak and Carbon Neutrality at the national level. Food waste is an essential part of organic waste. It's reported that under a recycle rate of 80%, the total carbon dioxide equivalent (CO_2^{-e}) emissions caused by sanitary landfill of organic waste can reduce approximately by 16% in Australia [5–7]. However, the recycle rate of food waste in China needs great improvement. In Shenzhen, the recycle rate of food waste from the residential areas and the catering industry were only 0.2% and 28% in 2019, respectively [4]. Thus, mapping the high-resolution distribution of food waste is of great significance, as it can provide references for generation amount estimation, collection route and strategy determination, and cascade facilities configuration of pretreatment and end-of-life treatment. Then, through establishing a comprehensive food waste treatment system, the food waste can be collected and treated in a more energy-saving and eco-friendly way. Thus, a comprehensive food waste treatment system can exploit the potentialities of reduction of greenhouse gas emissions during food waste recycle and disposal [8].

Xu et al. summarized a basic framework of municipal solid waste generation forecasting, and the core issues were concluded as scale and resolution, factor and method [9]. The prediction conducted under a larger scale and a higher resolution means the growth of the predicted units, and consequently, more data are required as input. Based on the identification of chosen scales and resolution, influencing factors are classified into intrinsic, socio-demographic, individual, natural and policy factors.

Mainstream methods of solid waste generation include regression analysis, time series analysis, grey system model, system dynamics and machine learning. Regression analysis is the most widely used traditional method, as it has simple calculation and can obtain satisfying accuracy by adjusting input parameters in the model, but it needs a large amount of data to screen proper input [10–13]. Shu et al. further predicted the spatial distribution of food waste production in Xiamen on the street scale based on regression analysis [14]. Autoregressive Integrated Moving Average (ARIMA) model is the most commonly used time series analysis, as it owns good accuracy in short-time prediction, but lacks interpretability in inner connections [15–18]. Grey system model was found in 1982 [19], and first applied in waste generation prediction in 2000 [20]. Grey system model is proven to have good performance under a lack of data, but it can only be applied in the common scenes under a few limited conditions [5, 21–24]. System dynamics can capture dynamic change, and has the advantage of simulating the inner connections of complicated system, but it's difficult to build the model and perfect the inner connections [25–28]. Considering that Machine learning (ML) has better performance when facing the complex non-linear problems, and it can obtain high accuracy by automatic parameters adjustment which saves time and manpower, in this study ML is applied.

ML is now applied in the prediction of waste generation, treatment and recycling processes. Guo et al. showed that the majority (54%) of relevant studies on organic waste adopted Artificial Neural Network (ANN) for modeling and prediction, followed by Support Vector Machine (SVM), Genetic Algorithm (GA), and Decision Tree (DT)/Random Forest (RF) [29]. ANN has successfully been applied to predict domestic, commercial, industrial, medical, and construction waste [30–33]. RF has strong interpretability and the ability to model the non-linear relationship in waste generation [34–36]. Other methods were also widely used in the prediction of various waste [37, 38].

To date, the distribution of waste generation at the city level has still not been comprehensively investigated. Current researches mainly focus on directly predicting waste generation at the city level using data released by official statistics agency, as the input data can be acquired in statistics and bulletin conveniently [30–38]. Few researches have completed the bottom-up waste generation prediction, by studying the waste generation of each emission unit, and aggregating the amount of waste generation of each emission unit. Also, the "black boxes" feature of ANN causes poor interpretability despite high accuracy [34]. Consequently, existing researches only can predict the total amount of waste generation, but it lacks analysis on the distribution of food waste. Either, though the

prediction result can acquire good accuracy, it can't effectively point out the denominating factor.

The main aim of this study is to predict FWG from the catering industry of Dongguan under high-resolution based on big data analysis and machine learning, and to provide suggestions and insights for food waste management and treatment. In this study, to the best of our knowledge, high-resolution prediction of FWG from the catering industry is first conducted. To acquire accuracy and interpretability at the same time, this study develops a machine learning method based on field survey data and Point of Interest (POI) data. Back Propagation Network (BPN) is applied to build the subsidiary FWG simulation model, which can provide references for interval setting, and RF is used to construct the FWG prediction model, which can directly predict the FWG of each POI. By predicting the waste generation of numerous interested points, regional waste generation can be realized through spatial analysis, and the prediction of waste distribution is completed in the scale of each interested object at the same time, then the resolution of the map can reach 1 km × 1 km.

The paper unfolds as follows: Sect. "Materials" briefly introduces the basic situation of Dongguan and data sources. The method adopted to FWG prediction and results visualization are described in Sect. "Method". Sect. "Results and discussion" explains the results, suggestions and insights based on the results are given. Conclusions are given in Sect. "Conclusion".

Materials

Study area

With a permanent resident population of over 10.47 million people [39], Dongguan has officially entered the ranks of the 14 megacities in China. As Dongguan connects Guangzhou with Shenzhen, Dongguan is considered the node city of the Guangdong–Hong Kong–Macao Greater Bay Area. Dongguan owns 32 sub-district administrative regions, which are divided into six functional areas, and the designed function for Shuixiang New Town Area (SNTA), Urban Area (UA), East Industrial Area (EIA), Coastal Area (CA), Songshanhu Area (SA) and Southeast Shenzhen-adjacent Area (SSA) is the zone for agriculture transformation and upgrading, city code enhancement, industrial and urban coordinated development, external communication leading, high-tech innovation, and Shenzhen spillover resources receiving, respectively (See Fig. 1) [39]. Table S1 listed the official statistics from the Dongguan Statistical Yearbook used by this study, and all the figures are

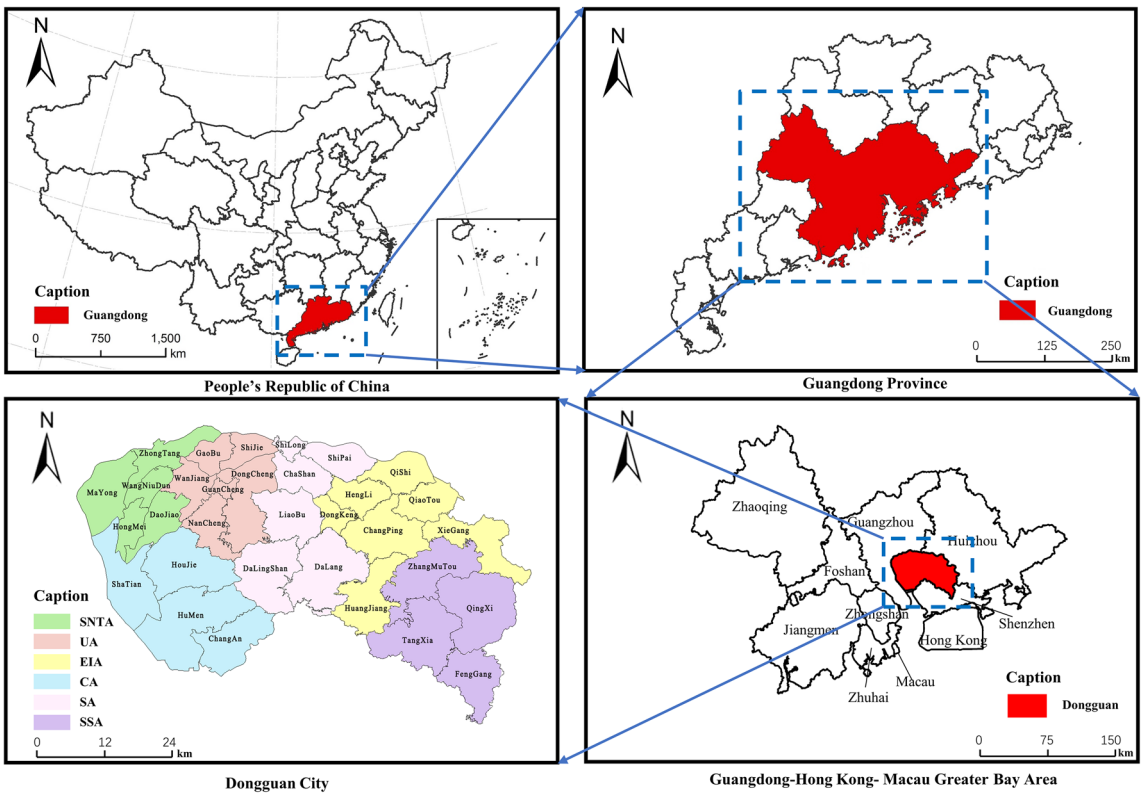


Fig. 1 Administrative map of Dongguan

graphed based on it [39]. According to the Dongguan Statistical Yearbook, the generation of domestic waste in Dongguan reached 5 million tons, and over 120,000 catering enterprises were registered in 2021 [39], and both the two mentioned indexes are expected to grow rapidly in the further future. The local city government has conducted a 3-year plan aimed at municipal waste classification, and will constantly improve the daily food waste treatment capacity [40].

Big data sources

The population kernel density map was collected from the Baidu map in June 2021 using python 3.7.1. The POI data was collected from Automap in June 2021, and we collected the geo-location (latitude and longitude) of over 96,000 restaurants in Dongguan. Then, the expected consumption expenditure per capita and seat number of these restaurants were filled based on the data from Dianping.com and a manual survey.

Survey data sources

From January to August 2021, the survey has been conducted in Dongguan. Thirty well-trained environmental volunteers, which were local college students, were dispatched to carry out questionnaire investigation in different functional regions in 6 areas of Dongguan. The survey questionnaire is attached as the Supplementary Material. A total of 879 restaurants were investigated, and the samples covered all 11 categories of restaurants according to the acquired POI data. The waste generation data was acquired from the 10-consecutive-day record in the ledger of food waste collection. To obtain official statistics data and acknowledge the existing problems in food waste treatment, the interview with local qualified enterprises of food waste treatment, the Leading Group for Waste Classification of Dongguan and local catering industry associations were conducted.

Except for the total FWG (FWG-T), the survey data can be divided into two categories. The first is static data, which reflects the basic situation of the restaurant, such as the restaurant type reflected by food waste production per capita (FWG-PC), seat numbers, consumption expenditure per capita (C-PC), and local consumption in the catering industry per capita (LCPC). Usually, the static data is determined in the beginning, and seldom changes throughout the whole business life. Another is dynamic data, it includes service count, turnover, and raw materials, which respectively correspond to consumer flow, fund flow, and material flow, and the difference in dynamic data between workday and off day can reflect the change in restaurant operation. Dynamic data can be used to simulate how a restaurant performs under different seasons, and static data can be used for the approximate prediction of different restaurants.

Besides, a typical Chinese food restaurant, which owned a seat number of 300, a C-PC of about 75 CNY and a service count of about 400 per day, was chosen to conduct the depth data collection. As the number of Chinese food restaurants occupies the largest share of total restaurant counts, it is of great importance to trace its material flow.

Methods

Principle of input parameters selection

Intrinsic factors are used the most widely, as it includes quantitative indexes in fields of economics, urbanization, and living, which can be measured directly and available from the official documents (statistics, bulletin, etc.). On the contrary, socio-demographic, individual, natural, and policy factors are qualitative indexes at most times. Researchers need to quantify the indexes based on given circumstances, and it brings unavailable personal errors. In comparison, intrinsic factors can most directly reflect the situation of waste generation. Consequently, studies on waste generation prediction should concentrate on intrinsic factors as prerequisites, and consider the spontaneous correction brought by socio-demographic and individual factors, and mandatory correction brought by natural and policy factors.

The correlation analysis can reflect the impacts of other survey data, which is divided into static data and dynamic data on total FWG (FWG-T), and the Correlation Coefficient (CC) is calculated as Eq. (1)

$$CC(X,Y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \sum (y-\bar{y})^2}} \quad (1)$$

where $\bar{x}$ and $\bar{y}$ $\bar{y}$ is the average of X and Y, respectively.

Table 1 shows the results of the correlation analysis. In terms of dynamic data, the ranking of impact on FWG-T is material flow, fund flow, and consumer flow, from the highest to the lowest. In terms of static data, the rankings are seat numbers, consumption expenditure per capita

Table 1 Correlation analysis of dynamic data and static data to FWG-T

	Dynamic data					Static data			
	Service count	Turnover	Meat	Vegetable	Others	Seat Number	FWG-PC	C-PC	LC-PC
FWG-T	0.28	0.32	0.41	0.49	0.88	0.32	0.081	0.35	−0.07

(C-PC), FWG-PC, and local consumption expenditure per capita (LC-PC) from the highest to the lowest. Compared with dynamic data, static data have better accuracy and stability in predicting FWG-T of different types of restaurants. Static data reflects the basic situation of the restaurant, and seldom changes throughout the whole business life. Dynamic data, on the one hand, lacks specific official systematic statistics, and on the other hand, it has a wide fluctuation range. Thus, dynamic data can be used into acquiring the distribution of FWG of the restaurant in different scenes, and based on the distribution, static data can be used into predicting the numerical interval of FWG.

Back propagation network (BPN) and random forest (RF)

BPN is a multi-layer forward neural ANN (See Fig. 2a). The weights of BPN are set randomly in the beginning, and continuously modified in the back propagation, until the differences between the training data and output data meet the training precision [41]. BPN computes the final value of the

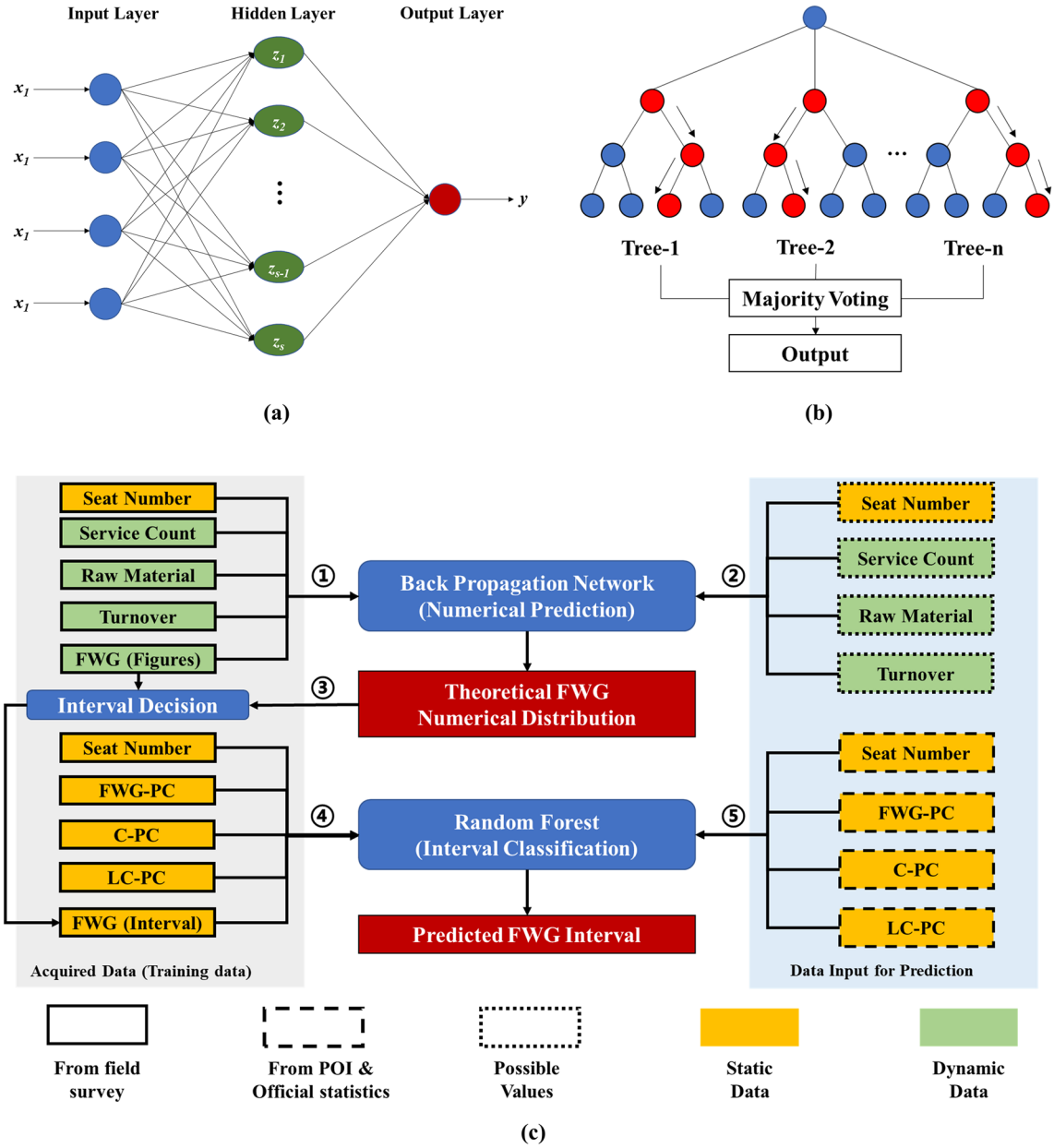


Fig. 2 Diagram of the used machine learning methods **a** The Schematic diagram of BPN; **b** The Schematic diagram of RF; **c** The structure of the used machine learning methods

hidden layer as shown in the Eq. (2) [42]. The decisive factor (R^2) and the mean absolute percentage error (MAPE) (See Eqs. (3) and (4)) are often used to evaluate the accuracy of ANN [43].

$$y = b + \sum x_i w_i \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - y_i)^2}{\sum_{i=1}^{n} (y_i - y_i^2} \tag{3}$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - y_i}{y_i} \right| \tag{4}$$

where b is bias, x_i is the initial variable value, w is weight, y_i is the actual value, and y_i is the predicted value.

RF is an ensemble learning method (See Fig. 2b). After leveraging the strategy of bagging and random feature, RF constructs several de-correlated decision trees and outputs their average predictions [44]. Overall accuracy (OA) is the proportion of correctly predicted samples to all samples [45], calculated as Eq. (5).

$$OA = \frac{True\ Positive + True\ Negative}{True\ Positive + True\ Negative + False\ Positive + False\ Negative} \tag{5}$$

The structure of the applied machine learning methods is shown in Fig. 2(c). Because the static data is collected at the specific interval, rather than a specific value, the problem of the numerical prediction of FWG turns into the interval estimation problem, which can be solved using a classifier in the artificial network. Thus, using proper intervals in the classification of FWG situations can both improve the prediction accuracy and the convergence rate of the artificial network. BPN is applied to simulate the FWG under different dynamic data. The model is trained by the dynamic data acquired from the field survey, using seat number, service count, turnover and raw material as input, and using FWG as output. All the input and output data are used in the form of specific values. Then a presumed list of dynamic data, which is considered to fully cover the indexes of the existing restaurants (See Table 2), is input to the model to acquire the frequency distribution of the generation of food waste. Based on the distribution, the proper interval size of FWG can be determined. RF is unexcelled in accuracy among current algorithms, and it estimates which variables are important for the classification. The survey data after being processed in the determined interval size are used to train the model, and the static data from POI are input into the model to predict the FWG of the restaurants in the POI. The output FWG of the RF model is in the form of classified intervals. Eventually, a numerical prediction problem is turned into an interval estimation problem.

Table 2 Chosen interval values of seat number and dynamic data

Seat number	Service count	Turnover (¥)	Raw material (kg)
50	250	2000	100
150	500	4000	200
250	750	6000	300
350	1000	8000	400
450	1250	10,000	500
550	1500	12,000	600
650	1750	14,000	700
750	2000	16,000	800
850	2250	18,000	900
950	2500	20,000	1000
1050	2750	22,000	1100
	3000	24,000	1200
	3250	26,000	1300
	3500	28,000	1400
	3750	30,000	1500
	4000	32,000	1600
	4250	34,000	1700
	4500	36,000	1800
	4750	38,000	1900
	5000	40,000	2000

Graphing and mapping

After acquiring the FWG of the restaurants in the POI, each restaurant is located based on its geographic information. An urban grid under the resolution of 1 km × 1 km is constructed, and the FWG of one grid is the total of the FWG of the restaurants in this grid. The flow map is drawn by e!Sankey software (version 4.5.2). The generation of food waste is counted into 1 km × 1 km grids, and then aggregated into grid levels. The spatial patterns are visualized by ArcGIS software (version 10.7).

Results and discussion

Variation of FWG in different categories of restaurants

Survey data analysis

Restaurants are divided into 11 categories as the same as the POI data, including Chinese food, Chinese fast food, western food, western fast food, bakery product, dessert, beverage, hot pot, Southeastern Cuisine, Japanese and Korean cuisine, and Hong Kong style cuisine. These 11 categories can basically cover all kinds of restaurants in China. Based on the

survey data, the FWG per capita (FWG-PC) of each category of the restaurant can be calculated. The results are compared with another sampling survey conducted in Chengdu, Sichuan Province [46] (See Table 3), as validation. The survey in this study lacks the data on buffet, but adds the data of other 6 categories of restaurant mentioned. The FWG-PC of Chinese food and hot pot are lower than Yang's survey, the reason is the difference in dietary habits between Dongguan and Chengdu, as Guangdong cuisine prefers a light diet and Sichuan cuisine prefers food using more oil and spicy sauce, especially in hot pot.

Suggestion for improvement of food waste collection standard

Separate collection and recycling of food waste is not only the integral terminal part of restaurant food waste management [47], but also the important beginning part of the following municipal food waste disposal system. Thus, it is of great importance to regulate the collection and recycling of food waste. Current Standard Technical Code for Food Waste Treatment has two versions in China. One is released by the Ministry of Housing and Urban–Rural Development of the People's Republic of China (CJJ 184–2012) [48] as industry standard, and the other is released by the Market Supervision Administration of Shenzhen Municipality (SZDBZ 252–2017) [49] as local standard. Both standards cover fields of definition, collection, transport, measurement, storage, and treatment, but lack emphasis on collection, especially in terms of the collection of different components. Current standards only mentioning the wasted oil should be collected and transported separately, as waste oil is considered the most valuable compound of food waste in recycling [50], and high oil content can cause clogging problems in food waste treatment facilities. As business operation modes in the catering industry are diversifying, the dynamic inventory of objects to collect from and content to collect is needed to be established, and guidelines for each type of object to deal with its food waste are also needed to be published. To comprehensively evaluate the recycling value of food waste, a measuring system should be innovated and standardized. Besides the weight, more parameters should be included, such as oil separation rate and purity. New parameters need new measuring methods and tools, and using artificial measurements to realize the rough measure of the component proportion in layered waste is one of the directions [51].

Table 3 Average Food waste generation per capita of each category of restaurant

Category	Survey data (kg)	Yang, et al. [46] (kg)
Chinese food	0.23	0.3–0.7
Western food	0.29	0.2–0.5
Fast food	0.13 Chinese 0.11 Western	0.1–0.2
Hot pot	0.24	0.5–1.0
Buffet	–	0.2–0.3
Beverage	0.09	–
Dessert	0.10	–
Japanese and Korean cuisine	0.85	–
Southeastern cuisine	0.69	–
Hong Kong style cuisine	0.21	–
Bakery	0.12	–

Material flow of FWG at the restaurant level

Material flow analysis

In order to identify key points of material flow during FWG, a Chinese food restaurant is chosen to carry out material flow analysis for seven consecutive days, and Fig. 3 shows the results. Raw materials consist of meat, vegetable, staple food, sauce, seafood, and others. Staple food includes rice, noodles, etc. The sauce includes oil, salt, etc. Others are the raw materials which are not included in the listed categories such as tofu. Food waste can be divided into avoidable and unavoidable wastes [52]. Unavoidable waste is the waste that can't be eaten and thrown away during cuisine due to the need for quality, such as bone, shell, and charred food. Avoidable waste is the waste that can be eaten but thrown away.

The quantification of food waste from the restaurant is based on weighting, waste composition estimation, and mass balance calculation [53]. The weight of raw material, leftover and food waste is measured by scales. After separating the liquid from the food waste, the waste composition estimation is conducted. The weight of avoidable waste is calculated as the weight of remaining food waste minus the weight of unavoidable waste. The weight of unavoidable waste is calculated based on the following facts: the proportion of meat and bone in poultry is 8:2, the proportion of meat and bone in livestock is 7:3, the proportion of meat and shell (or bone) in seafood is 9:1. The total weight of food waste is about 120 kg, and liquid takes the share of approximately 60 kg.

As it can be seen from Fig. 3, a main part of raw materials (86.7%) is delivered to be served after cuisine, and the rest (13.3%) is abandoned as leftover. Then, the served part is mainly consumed (53.8%), and the rest are wasted. The proportion of avoidable and unavoidable wastes is approximately 3:1. In a similar research conducted in school canteens in Malaysia, it also found that unavoidable waste was the major part of food waste [54]. It can be seen that the

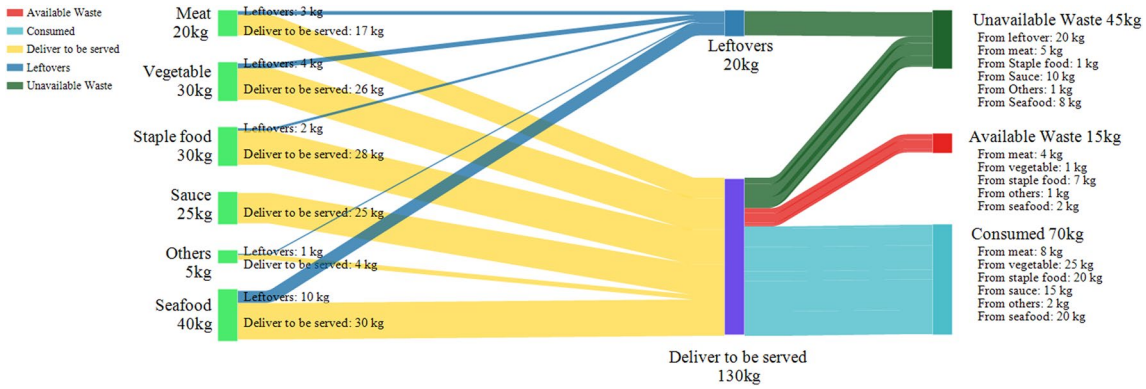


Fig. 3 Result of material flow of a Chinese food restaurant in one day

most direct way to reduce FWG in the restaurant is the reduction of avoidable waste generation, and the reduction of unavoidable waste generation can also be realized through the upstream process.

Suggestion for source reduction of FWG at the restaurant level

FWG exists in the whole food supply chain [55]. As restaurants can be considered as the downstream part of the food supply chain, FWG is determined by the supplying capacity and consuming capacity of raw materials. From the perspectives of material flow, the raw material is supplied by restaurants and consumed by consumers. As Fig. 3 shows, among two parts of food waste, unavoidable waste mainly relates to restaurants, and avoidable waste mainly relates to consumers. About 75% of food waste comes from unavoidable waste, and to be more specific, 20% comes from the leftover. To reduce leftover, restaurants can optimize the cooking process, simplify presentations or push out new dishes to fully use the raw material [56]. To reduce the rest of the unavoidable waste, purchasing pretreated raw materials that are free of uneatable parts can be a good way, so that this part of waste can be left in the supplier to wait for centralized treatment and eatable part of the delivered dishes becomes more. Also, the restaurant should reasonably make operating plans and scientifically prepare raw materials based on real needs, and it can avoid raw materials being discarded due to surplus, and proper reuse and storage of the remaining materials are also important. FWG in the food preparation part is thought to be huge [57], and the central kitchen system is a new way to meet the needs of material supply and food waste reduction in chain restaurants [58]. The food factory and the central kitchen produce semi-products based on the needs of restaurants, and semi-product can remain in a good status in the central kitchen for 4 to 5 days. Restaurants are asked to order raw materials for needs of 2 days each time, and can send orders to the central kitchen for real-time delivery based on actual needs. In this way, the shelf life of semi-products can fully recover the service life of restaurants, which makes sure all the semi-products can be consumed in 2 days. Restaurants can also play a role by offering a different quantity of meals to meet the needs of different number of consumers, and changing set-meal style into individual-meal style [59]. For the government, as the behavior of saving food waste has relied more on personal virtue than on politics or religion in China, the Clean Your Plate Campaign should be further continuously conducted within long-term school education and media publicity with appropriate incentives [60].

Patterns of FWG at the restaurant level

Patterns of FWG under different operation data

It can be noticed during the survey that the amount of seat number and dynamic data has a range, and to simulate all the situations of different restaurant operations, this study selects several interval values to make combinations as the input. The total number of combinations is 88000, and the chosen interval values are listed in Table 2. The R^2 and MAPE of the BPN model are 0.96 and 8.12%. After rounding the predicted FWG, the frequency of each value is counted (See Fig. (a). The curve can be divided into high-, middle-, and low-frequency zones based on the trend, and several peaks can be discovered in high- and middle-frequency zones. The value of the discovered peaks can be considered the value with the highest probability of occurrence among the surrounding values. Because the frequency of 0–15 kg is 32052, which is significantly higher, the first interval is set to be 0–15 kg. In high- and middle- frequency zones, the average interval of the peaks is 32.5 kg and 44.75 kg, respectively. For convenience, the mentioned interval is set to 30 kg and 45 kg, respectively. In the low-frequency zone, no obvious peak can be discovered, and we set the interval to 105 kg here. Above all, the intervals of 15 kg, 30 kg, 45 kg, and

105 kg are used in the division of intervals of 0–15 kg, 15–105 kg, 105–330 kg, and 330–2115 kg, respectively.

Figure 4b shows the frequency distribution when dividing the result into blocks of 15 kg. BPN simulation and survey data have a similar frequency distribution, in which the interval of 0–15 kg takes the largest share, then the second largest share is occupied by the intervals between 15 and 105 kg in the survey, and between 15 and 285 kg in BPN simulation. The total frequency of remaining results, which are between 285 and 2115 kg, is less than 4% in the survey, and less than 10% in the BPN simulation, and it suggests that the interval of 0–105 kg should be divided into narrower intervals, then the expanded interval can be used to divide the interval of 105–305 kg, and the division of the interval of 305–2115 kg can be even looser. Figure 4c shows the frequency distribution when using 15 kg, 30 kg, 45 kg, and 105 kg in the division of intervals of 0–15 kg, 15–105 kg, 105–330 kg, and 330–2115 kg, respectively. As it shows, the result basically concentrates on 0–330 kg, and is sparsely distributed in the remaining intervals. Then the number of intervals is smaller, and it reduces the number of dimensions when using the on-hot encoder method, and the fluctuation in each interval brings similar error to the basic FWG.

Figure 5 shows the variation trend between the dynamic data and FWG. The color and the size of the data point reflect the quantity of FWG, and the bigger the size is, the larger the quantity is. The relation between color and quantity can refer to the scale in the figure. It shows that FWG is highly increased with the growth of raw material under the same circumstance, and on the contrary, with the same amount of raw material, FWG decreases as the seat number or service count rises. This result has indicated that FWG shows a positive correlation with raw material supply capacity, and a negative correlation with raw material consume capacity. With other parameters remaining the same, FWG increases with the rising turnover, but not significant as with the rising seat number and service count. It suggests that an upscale restaurant tends to generate more food waste, as the consumption expenditure per capita is higher, which may relate to stricter material requirements and more complicated cuisine craft. The highest FWG happens under the situation

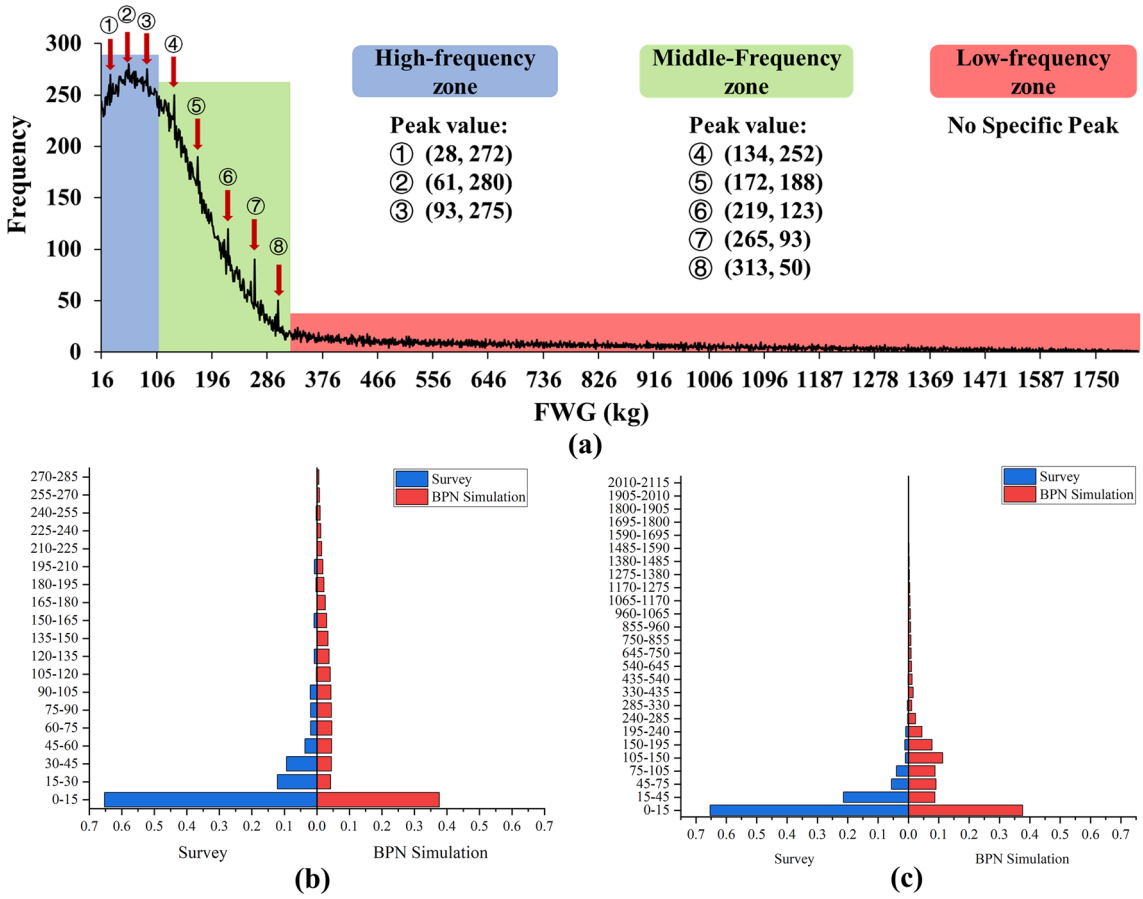


Fig. 4 Frequency distribution of the FWG based on different intervals **a** Based on intervals of 1 kg; **b** Based on equal intervals of 15 kg; **c** Based on progressive intervals

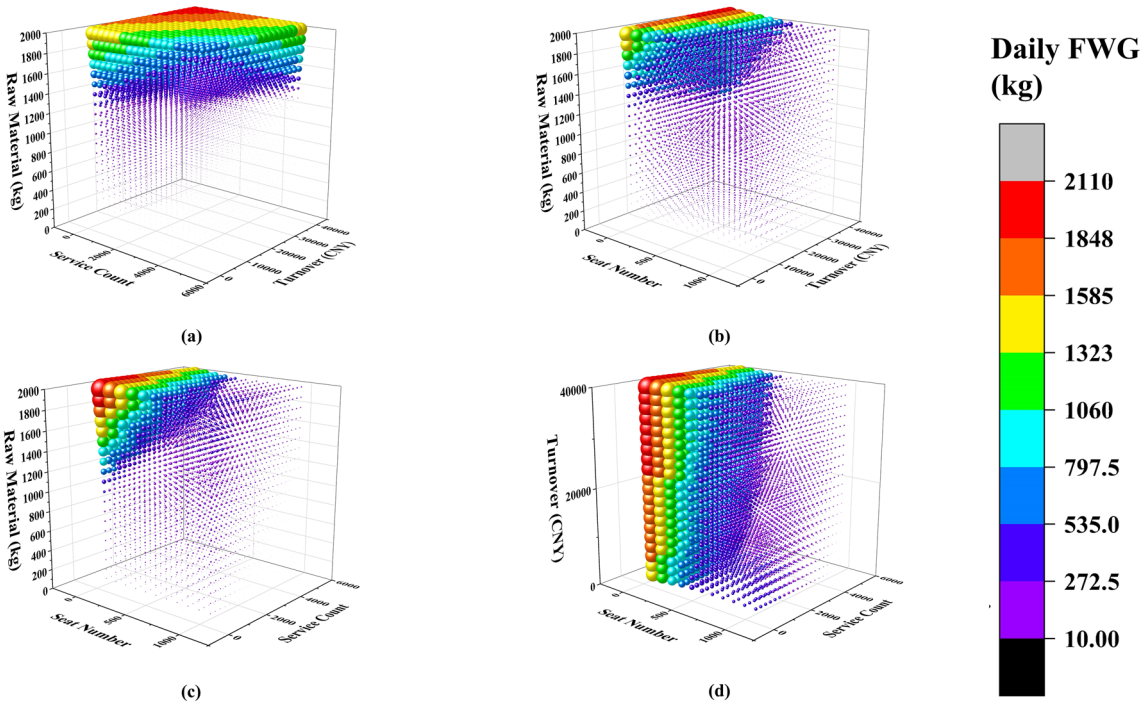


Fig. 5 Distribution of food waste as dynamic data change **a** Service count, turnover, and raw materials; **b** Seat Number, raw materials, and turnover; **c** Seat number, service count, and turnover; **d** Seat number, service count, and raw materials

of the highest raw material and consumption expenditure per capita, and the lowest seat number and service count.

Insights into food waste collection system

As further reliable food waste collection services can facilitate the timely food waste separation and recycling of the restaurant [47], the establishment of a food waste collection and transport information system (FWCTIS) should be put on the agenda. The survey in Dongguan showed that due to the information asymmetry between restaurants and qualified enterprises of food waste treatment, the problem of the food waste couldn't be collected by qualified enterprises in time, and treatment capacity can't be fully used (about 50% of the designed capacity) with surplus transport capacity in the same time. This results in misunderstanding, which causes negative effects, such as restaurants selling food waste to unqualified enterprises illegally, and qualified enterprises are not willing to provide service to restaurants far from the plant due to high transport cost. FWCTIS is designed to eliminate information asymmetry. Restaurants can upload their needs and demand on collecting frequency and time, and realize the online set-up of the ledger for collection. Qualified enterprises can in real time deploy collector vehicles based on the received information, and adjust the operation of the treatment plant. What's more, FWCTIS should have the ability to solve the collecting vehicle routing problem and the FWG predicting problem. The competent department should also have access to supervise the system. In a case conducted in Canada, using smart technology and geo-localization means to plan the food waste collection routes, based on 12.2 km between sites and a fleet of 200 trucks, a potential reduction of 940 t CO_2^{-e} per year can be realized [61].

Spatial Patterns of FWG at the city level

Distribution of FWG under high resolution

In this study, Random Forest Network is found to be the best prediction model, as it owns an OA of 0.85, and other models, such as the BPN and RBFN (radial basis function network), own a significantly lower OA of about 0.5. Operating a feature importance check is a necessary step before running a Random Forest Network, and the result of RF shows a different result compared with Fig. 4. The importance feature of seat number, C-PC, LC-PC, and FWG-PC is 0.42, 0.24, 0.21, and 0.13, respectively, and the importance of seat number and LC-PC has been improved, which might be related to the unique operation mechanism of Random Forest Network.

FWG patterns of 96,303 restaurants are predicted, and the predicted FWG from the catering industry in Dongguan is about 3106.00 t per day. The results are listed in Tables

S2–S5. According to the sample survey on the composition of domestic waste conducted in Guangzhou in 2021 [62], the share of food waste in domestic waste is 52.99%. Thus, it can be estimated that the FWG in Dongguan is about 7258.90 t per day, and the part from the catering industry takes the share of approximately 42.79%.

Seat number has the highest figure both in correlation analysis and feature importance analysis, indicating that as the seat number grows, share of count and generation both decrease. Small-sized and medium-sized restaurants, whose seat number is between 0 and 100, take the highest share of 93.75% in count number, and contribute over 90% of the total food waste. As a result, the frequency of the interval of 16–45 kg reaches 95.22%. Compared with the result of the prediction of BPN, the highest frequency in FWG is contributed by 16–45 kg rather than 0–15 kg, and the reason is that to generate 0–15 kg food waste, raw material supply capacity is needed to be strictly close with raw material consume capacity, which is hard to satisfy in reality. C-PC reflects the grades of the restaurants. The restaurants can be divided into the eatery, the middle-class restaurant, the upscale restaurant, and the luxury restaurant, whose C-PC is 0–50, 50–100, 100–300, and beyond 300, respectively. Eatery has the highest total and contributes the most food waste. The category of restaurant is also an important factor. Chinese fast food restaurants and Chinese food restaurants together take 38.15% and 47.37% of restaurant count, and FWG is in accordance with local Chinese diet habits [63]. It's noticeable that beverage shop ranks third in restaurant count and FWG and its fresh sugary beverage has a massive of potential consumers, such as milky tea and fruit tea. However, the moisture content of food waste generated in beverage shops is extremely higher than the others, because this food waste is mainly come from the remaining liquid and sugary leftovers. Also, the generation can have a large fluctuation, as the staff sometimes dump the liquid waste directly into the drain. The FWG of the bakery is also arguable, as the bakery often overproduces to satisfy the need for display, and the waste can be re-used or donated.

Figure 6 shows the distribution map of food waste, the catering industry, population, and turnover of catering industry. Figure 6(a, b) is graphed based on a metric fishnet of 1 km × 1 km, and Fig. 6(c) is vectorized based on the population thermogram collected from the Gaode map. At the grid level, the distribution of population, restaurant count, and FWG has a successive trend. The FWG in area level for UA, CA, SSA, SA, EIA, and SNTA was 22.37, 23.33, 14.49, 19.54, 15.00%, and 5.28%, respectively. High-density zones in UA and CA were linked together. In SSA, it formed three high-density zones in the middle of the three main towns. In EIA and SA, discrete high-density zones can be found, which may attribute to the sporadic urbanized regions in these two areas. SNTA only formed a middle-density zone in the restricted town center.

Figure 6d shows the correlation between the distribution of FWG, population, GDP, local catering industry income (LCII), and LCPC at the area level. The correlation coefficients of population-FWG, Restaurant count-FWG,

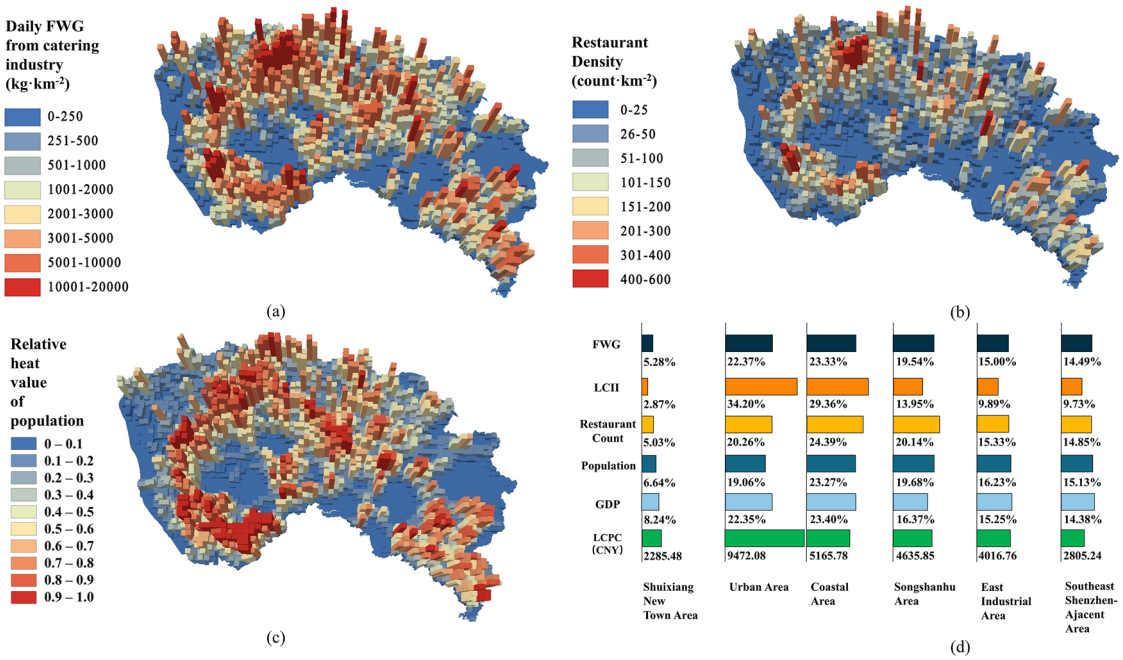


Fig. 6 Distribution map of food waste, restaurant, population and turnover of catering industry, **a** Food waste; **b** Restaurant; **c** Population; **d** Correlation;

GDP-FWG, LCII-FWG, and LCPC-FWG are 0.97, 0.99, 0.9, 0.88, and 0.74, respectively. LCII reflects the local catering business status, and LCPC represents the potential of local individual consumption. A high urbanization rate means a large population and a prosperous economic level. Due to the busier lifestyle and the stronger spending power of residents, the growth of the urbanization rate brings about an increasing number of restaurants and a more frequent eat-out rate [64]. It suggests that the urbanization level is a significant factor in the FWG from the catering industry, as the local population provides the base for the potential consumers, and local GDP determines the spending power of consumers. Food waste contains parts from the catering industry and the residential area, and it can also be extrapolated that the proportion of the food waste from the catering industry will increase with the growth of the urbanization rate.

Insights into distributed food waste treatment system

Distributed food waste treatment system is considered one of the best ways to solve the problem that food waste has a larger amount of generation and a wide range of distribution. According to Technical Code for Food Waste Treatment (CJJ 184–2012) [48], food waste treatment plant is divided into category I, II, III, and IV, with the daily treatment of beyond 300 t, 150–300 t, 50–150 t, and under 50 t, respectively. The local government of Dongguan is conducting a plan to construct category III and IV treatment plants in every town and street, even in community groups, to realize the real-time and in situ treatment of food waste. With existing plants with the largest capacity in Mayong (300 t/d) and Nancheng (200 t/d), Dongguan is building a distributed food waste treatment system, which uses anaerobic fermentation as centralized treatment, and uses high-temperature composting and biological transformation as decentralized treatment. If each treatment plant is conducting end-of-life treatment, then the capacity of the plant is limited by technological requirements, causing high economical costs and a huge number of plants to match the FWG. A high-resolution map can provide reference to the location, service scale, and capacity of the treatment plant in the distributed food waste treatment system, and is also beneficial to the route optimizing problem.

Thus, the distributed food waste treatment system should consist of pre-treatment and end-of-life treatment facilities to realize cascade utilization [8]. From the perspectives of the administrative precinct, it is estimated that the FWG of towns and streets in Dongguan is between 72 t/d and 1432 t/d. For districts with small FWG, adjacent regions can combine to jointly construct an end-of-life plant, or districts with large FWG can expand their capacity to receive the waste which is pretreated in adjacent regions. From the perspectives of cascade utilization, the facility with primary treatment has a larger capacity. Wen et al. [65] found that limited processing capacity of waste disposal facilities and low rates of sorting waste cause low efficiency of the whole food waste treatment system. Using primary treatment facilities to separate oil and moisture can significantly reduce the weight but improve the purity of food waste in the end-of-life treatment, and it can solve the problem of the lack of end-of-life treatment capacity and transport capacity. Another reason of in-community groups constructing primary treatment facility is better than end-of-life facility in-community groups is that end-of-life product may not be local consumed. A centralized processing can centralize the use of product and improve efficiency. When choosing technological routes of facility, local government should comprehensively consider the cost, profit, and especially the market-consuming power of the product, to avoid the recycling product from becomes waste again [66].

Conclusion

This study has first developed a machine learning method to predict the FWG from the catering industry under high-resolution based on grabbed POI data and filed survey data. In this method, a BPN model was used to predict FWG under different operation data. Then, based on the acquired theoretical FWG numerical distribution, the intervals used to divide FWG values were determined. By this step, the numerical prediction issue has been transformed into the interval classification issue, and eventually, a RF model was used to predict the FWG intervals of the restaurants in the POI data. FWG patterns of 96,303 restaurants are predicted, and the predicted FWG from the catering industry is about 3,106.00 t per day. Variations of FWG in different categories of restaurants, the material flow of FWG at the restaurant level, patterns of FWG at the restaurant level, and spatial patterns of FWG at the city level were investigated. Based on the mentioned findings, suggestions for improvement of food waste collection standard and source reduction of FWG, and insights into the food waste collection system and distributed food waste treatment system were raised.

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Declarations

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References

1. National Bureau of Statics (2021) Bulletin of the seventh National Census. http://www.stats.gov.cn/tjsj/tjgb/rkpcgb/qgrkpcgb/
2. Ministry of Housing and Urban-Rural Development of the People's Republic of China (2020) Urban Construction Statistical Yearbook 2019. http://www.mohurd.gov.cn/xytj/tjzljsxytjgb/jstjnj/w020201231224852714231250000.xls
3. Ministry of Housing and Urban-Rural Development of the People's Republic of China (2017) Domestic Waste Classification Regulation. http://www.mohurd.gov.cn/wjfb/201906/t20190606_240787.html
4. Wang H, Li H, Yin M, Du X, Zhou Y, Li T, Liu J (2020) Analysis on source emission rule and resource recovery path of MSW in Shenzhen. Environ Sanitat Eng 28(3):21–27. https://doi.org/10.19841/j.cnki.hjwsgc.2020.03.004
5. Kiran M, Shanmugam PV, Mishra A, Mehendale A, Sherin NHR (2021) A multivariate discrete grey model for estimating the waste from mobile phones televisions and personal computers in India. J Cleaner Prod. https://doi.org/10.1016/j.jclepro.2021.126185
6. Australian Organics Recycling Association (2020) Australian Organics Recycling Industry Capacity Assessment: 2020–21. Department of Agriculture, Water and the Environment of Australia, Green Industries South Australia. https://www.environment.gov.au/system/files/resources/fa3f70c8-44d2-42e6-b57d-0a94114832ec/files/australian-organics-recycling-industry-capacity-assessment-2020-21.pdf
7. Blue Environment (2020) National Waste Report 2020. Department of Agriculture, Water and the Environment. https://www.environment.gov.au/system/files/pages/5a160ae2-d3a9-480e-9344-4eac42ef9001/files/national-waste-report-2020.pdf
8. Ascher S, Li W, You S (2020) Life cycle assessment and net present worth analysis of a community-based food waste treatment system. Bioresour Technol. https://doi.org/10.1016/j.biortech.2020.123076
9. Xu L, Cui S, Yan Z, Lin T (2013) Core issues of forecasting municipal solid waste generation. Environ Sci Technol 32(2):184–190. https://doi.org/10.3969/j.issn.1003-6504.2013.02.038
10. Winer AM, Peters JW, Smith JP, Pitts JN (2002) Response of commercial chemiluminescent nitric oxide-nitrogen dioxide analyzers to other nitrogen-containing compounds. Environ Sci Tech 8(13):1118–1121. https://doi.org/10.1021/es60098a004
11. Benitez SO, Lozano-Olvera G, Morelos RA, Vega CA (2008) Mathematical modeling to predict residential solid waste generation. Waste Manag 28(Suppl 1):S7–S13. https://doi.org/10.1016/j.wasman.2008.03.020
12. Ghinea C, Dragoi EN, Comanita ED, Gavrilescu M, Campean T, Curteanu S, Gavrilescu M (2016) Forecasting municipal solid waste generation using prognostic tools and regression analysis. J Environ Manage 182:80–93. https://doi.org/10.1016/j.jenvman.2016.07.026
13. Al-Salem SM, Al-Nasser A, Al-Dhafeeri AT (2018) Multi-variable regression analysis for the solid waste generation in the State of Kuwait. Process Safety Environ Protect 119:172–180. https://doi.org/10.1016/j.psep.2018.07.017
14. Shu S, Wang R, Sun Y, Liu J (2012) Spatial down-scaling analysis and simulation of kitchen waste density. Environ Sci Technol 35(61):458–463
15. Lin J-g (1998) Feasibility study on prediction of properties of municipal solid waste with time series models. J Hazardous Materials 58(1–3):47–57. https://doi.org/10.1016/s0304-3894(97)00119-2
16. Navarro-Esbrí J, Diamadopoulos E, Ginestar D, (2002) Time series analysis and forecasting techniques for municipal solid waste management. Resources, Conservation Recycling 35(3):201–214. https://doi.org/10.1016/s0921-3449(02)00002-2
17. Wu L, Lu J, Liao L, Jiang J (2013) Prediction of domestic waste output based on ARIMA model. Environ Sanitation Eng 21(5):1–4. https://doi.org/10.3969/j.issn.1005-8206.2013.05.001
18. Song Z, Li Y, Huang Z (2015) Analysis and Forecast of Construction Waste Based on ARIMA Model. 5th International Conference on Civil Engineering and Transportation; Guangzhou.
19. Ju-Long D (1982) Control problems of grey systems. Sys Control Lett 1(5):288–294. https://doi.org/10.1016/s0167-6911(82)80025-x
20. Chen HW, Chang NB (2000) Prediction analysis of solid waste generation based on grey fuzzy dynamic modeling. Resour Conservation Recycling 29(1–2):1–18. https://doi.org/10.1016/S0921-3449(99)00052-X
21. Liu G, Yu J (2007) Gray correlation analysis and prediction models of living refuse generation in Shanghai city. Waste Manag 27(3):345–351. https://doi.org/10.1016/j.wasman.2006.03.010
22. Zhao M, Zhao C, Yu LL, Li GM, Huang JW, Zhu HC, He WZ (2016) Prediction and analysis of WEEE in China based on the gray model. Selected Proceed Tenth International Conference Waste Manag Technol 31:925–934. https://doi.org/10.1016/j.proenv.2016.02.113
23. Duman GM, Kongar E, Gupta SM (2019) Estimation of electronic waste using optimized multivariate grey models. Waste Manag 95:241–249. https://doi.org/10.1016/j.wasman.2019.06.023
24. Zhou W, Zeng B, Wang J, Luo X, Liu X (2021) Forecasting Chinese carbon emissions using a novel grey rolling prediction model. Chaos Solitons Fractals. https://doi.org/10.1016/j.chaos.2021.110968
25. Tseng C-H, Hsu Y-C, Chen Y-C (2019) System dynamics modeling of waste management, greenhouse gas emissions, and environmental costs from convenience stores. J Cleaner Production. https://doi.org/10.1016/j.jclepro.2019.118006
26. Zhu C, Fan R, Luo M, Lin J, Zhang Y (2020) Urban food waste management with multi-agent participation: a combination of evolutionary game and system dynamics approach. J Cleaner Production. https://doi.org/10.1016/j.jclepro.2020.123937
27. Galeano DAR, Rodriguez SCB (2021) An integrated method of environmental analysis and system dynamics for management of mobile phone waste in Colombia. J Cleaner Production. https://doi.org/10.1016/j.jclepro.2020.123768
28. Rafew SM, Rafizul IM (2021) Application of system dynamics model for municipal solid waste management in Khulna city of Bangladesh. Waste Manag 129:1–19. https://doi.org/10.1016/j.wasman.2021.04.059
29. Guo H, Wu S, Tian Y, Zhang J, Liu H (2021) Application of machine learning methods for the prediction of organic solid waste treatment and recycling processes: a review. Bioresour Technol. https://doi.org/10.1016/j.biortech.2020.124114

30. Tiwari MK, Samir Bajpai UD (2012) Prediction of industrial solid Waste with ANFIS model and its comparison with ANN Model-a case study of Durg-Bhilai Twin City India. Internat J Eng Innovative Technol 9001:192–201
31. Golbaz S, Nabizadeh R, Sajadi HS (2019) Comparative study of predicting hospital solid waste generation using multiple linear regression and artificial intelligence. J Environ Health Sci Eng 17(1):41–51. https://doi.org/10.1007/s40201-018-00324-z
32. Coskuner G, Jassim MS, Zontul M, Karateke S (2021) Application of artificial intelligence neural network modeling to predict the generation of domestic, commercial and construction wastes. Waste Manag Res 39(3):499–507. https://doi.org/10.1177/0734242X20935181
33. Lu W, Lou J, Webster C, Xue F, Bao Z, Chi B (2021) Estimating construction waste generation in the greater bay area. China Using Machine Learning Waste Manag 134:78–88. https://doi.org/10.1016/j.wasman.2021.08.012
34. Kannangara M, Dua R, Ahmadi L, Bensebaa F (2018) Modeling and prediction of regional municipal solid waste generation and diversion in Canada using machine learning approaches. Waste Manag 74:3–15. https://doi.org/10.1016/j.wasman.2017.11.057
35. Solano Meza JK, Orjuela Yepes D, Rodrigo-Ilarri J, Cassiraga E (2019) Predictive analysis of urban waste generation for the city of Bogota, colombia, through the implementation of decision trees-based machine learning, support vector machines and artificial neural networks. Heliyon. https://doi.org/10.1016/j.heliyon.2019.e02810
36. Rutqvist D, Kleyko D, Blomstedt F (2020) An automated machine learning approach for smart waste management systems. Ieee Transactions Indust Informatics 16(1):384–392. https://doi.org/10.1109/Tii.2019.2915572
37. Ferreira JA, Costa M, Tereso A, Oliveira JA (2015) A multi-criteria decision support system for a routing problem in waste collection. International Conference on Evolutionary Multi-Criterion Optimization
38. Hu R, Chen K, Chen W, Wang Q, Luo H (2021) Estimation of construction waste generation based on an improved on-site measurement and SVM-based prediction model: a case of commercial buildings in China. Waste Manag 126:791–799. https://doi.org/10.1016/j.wasman.2021.04.012
39. Bureau of Statistics of Dongguan (2022) Dongguan Statiscal Yearbook 2021. http://tjj.dg.gov.cn/tjnj/index.html
40. The People's Government of Dongguan City in Guangdong Province (2020) Three-year Action Plan of Municipal Solid Waste Classification in Dongguan (2020–2022). http://www.dg.gov.cn/gkmlpt/content/3/3169/post_3169971.html#684
41. Wu D, Zhang D, Liu S, Jin Z, Chowwanonthapunya T, Gao J, Li X (2020) Prediction of polycarbonate degradation in natural atmospheric environment of China based on BP-ANN model with screened environmental factors. Chem Eng J. https://doi.org/10.1016/j.cej.2020.125878
42. Suparmaniam U, Shaik NB, Lam MK, Lim JW, Uemura Y, Shuit SH, Show PL, Tan IS, Lee KT (2022) Valorization of fish bone waste as novel bioflocculant for rapid microalgae harvesting: Experimental evaluation and modelling using back propagation artificial neural network. J Water Process Eng. https://doi.org/10.1016/j.jwpe.2022.102808
43. Jiang Y, Huang J, Luo W, Chen K, Yu W, Zhang W, Huang C, Yang J, Huang Y (2023) Prediction for odor gas generation from domestic waste based on machine learning. Waste Manag 156:264–271. https://doi.org/10.1016/j.wasman.2022.12.006
44. You H, Ma Z, Tang Y, Wang Y, Yan J, Ni M, Cen K, Huang Q (2017) Comparison of ANN (MLP), ANFIS, SVM, and RF models for the online classification of heating value of burning municipal solid waste in circulating fluidized bed incinerators. Waste Manag 68:186–197. https://doi.org/10.1016/j.wasman.2017.03.044
45. Xiong N, Lu H, Yang X, Wang J, Yue D (2022) Spatial characteristics and multifactorial driving analysis of fly-tipping bulky waste in Beijing based on the random forest model. J Cleaner Production. https://doi.org/10.1016/j.jclepro.2022.132534
46. Yang S, Wang K, Wu S, Zhou Y (2018) Prediction on generation quantity of food waste in planning of kitchen waste disposal facilities. Environ Sanitation Eng 26(003):89–90
47. Filimonau V, De Coteau DA (2019) Food waste management in hospitality operations: a critical review. Tour Manage 71:234–245. https://doi.org/10.1016/j.tourman.2018.10.009
48. Ministry of Housing and Urban-Rural Development of the People's Republic of China (2012) Technical Code for Food Waste Treatment. China Architecture & Building Press, Beijing
49. Market Supervision Administration of Shenzhen Municipality (2017) Technical Code for Food Waste Treatment. Shenzhen city, Guangdong Province.
50. Lang L, Wang Y, Chen X, Zhang Z, Yang N, Xue B, Han W (2020) Awareness of food waste recycling in restaurants: evidence from China. Resources, Conservation Recyclin. https://doi.org/10.1016/j.resconrec.2020.104949
51. Cropley M, Sprajcer M, Dawson D (2022) Wastogram: Validation of a new tool to measure household food waste. J Environ Psy. https://doi.org/10.1016/j.jenvp.2022.101896
52. Papargyropoulou E, Lozano RK, Steinberger J, Wright N, Ujang ZB (2014) The food waste hierarchy as a framework for the management of food surplus and food waste. J Cleaner Product. https://doi.org/10.1016/j.jclepro.2014.04.020
53. de Oliveira PT, da Silva CA, Conejero MA, Rodrigues Deliberador L, Otávio Batalha M (2022) Food waste measurement in a chain of industrial restaurants in Brazil. J Cleaner Production. https://doi.org/10.1016/j.jclepro.2022.133351
54. Kasavan S, Ali NIBM, Ali SSBS, Masarudin NAB, Yusoff SB (2021) Quantification of food waste in school canteens: a mass flow analysis. Resour, Conservat Recycling. https://doi.org/10.1016/j.resconrec.2020.105176
55. Lipinski B. HC, Lomax J., Kitinoja L., Waite R., Searchinger T. (2013) World Resources Institute Working Paper.1–40.
56. Spang ES, Moreno LC, Pace SA, Achmon Y, Donis-Gonzalez I, Gosliner WA, Jablonski-Sheffield MP, Momin MA, Quested TE, Winans KS, Tomich TP (2019) Food loss and waste: measurement, drivers, and solutions. Annual Rev Environ Resour 44(1):117–156. https://doi.org/10.1146/annurev-environ-101718-033228
57. SRA (2010) Too Good to Waste: Restaurant Food Waste Survey Report. London: SRA (Sustainable Restaurant Association).
58. Gao S, Bao J, Li R, Liu X, Wu C (2021) Drivers and reduction solutions of food waste in the Chinese food service business. Sustainable Production Consumption 26:78–88. https://doi.org/10.1016/j.spc.2020.09.013
59. Heikkilä L, Reinikainen A, Katajajuuri J-M, Silvennoinen K, Hartikainen H (2016) Elements affecting food waste in the food service sector. Waste Manage 56:446–453. https://doi.org/10.1016/j.wasman.2016.06.019
60. Wang L, Yang Y, Wang G (2022) The clean your plate campaign: resisting table food waste in an unstable world. Sustainability. https://doi.org/10.3390/su14084699
61. Montecinos J, Ouhimmou M, Chauhan S, Paquet M (2018) Forecasting multiple waste collecting sites for the agro-food industry. J Cleaner Production 187:932–939. https://doi.org/10.1016/j.jclepro.2018.03.127
62. Bureau GUMaCLE (2022) Announcement of Guangzhou Municipal Bureau of Urban Management and Comprehensive Law Enforcement on the results of the sampling survey on the

composition and nature of domestic waste in Guangzhou in 2021. http://cg.gz.gov.cn/zwgk/tzgg/content/post_8186588.html
63. Song G, Semakula HM, Fullana-i-Palmer P (2018) Chinese household food waste and its' climatic burden driven by urbanization: a bayesian belief Network modelling for reduction possibilities in the context of global efforts. J Cleaner Product 202:916–924. https://doi.org/10.1016/j.jclepro.2018.08.233
64. Ogunmoroti A, Liu M, Li M, Liu W (2022) Unraveling the environmental impact of current and future food waste and its management in Chinese provinces. Resources, Environment and Sustainability. https://doi.org/10.1016/j.resenv.2022.100064
65. Wen Z, Wang Y, De Clercq D (2015) Performance evaluation model of a pilot food waste collection system in Suzhou City. China J Environ Manage 154:201–207. https://doi.org/10.1016/j.jenvman.2015.02.025
66. Fang X, Gao B, Zhong D, Wang L, Borrion A, Huang W, Xu S, Cui S (2023) Closing the food waste loop: Analysis of the agronomic performance and potential of food waste disposal products. J Cleaner Product. https://doi.org/10.1016/j.jclepro.2022.135174

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